

Factor momentum in commodity futures markets

Yiyan Qian*

Ying Jiang[†]

Xiaoquan Liu[‡]

This version: February 5, 2024

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This paper examines the factor momentum in commodity futures markets. Based on the US and UK data from 1985 to 2022, we first show that a commodity factor's past returns positively predict its future returns. This predictability leads to sizable economic profits in a long-short trading strategy, is at its strongest over the one-month horizon, and could be explained by mispricing. The factor momentum suggests mean-variance inefficient commodity factors and negatively impacts the efficiency of pricing models. We then construct the time-series efficient factors, which exhibit higher Sharpe ratios and improve the performance of pricing models. These findings are robust across international commodity futures markets. Our results point to the potential to time commodity factors and highlight the importance of conditional asset pricing in commodity futures markets.

JEL classification: G11, G12.

Keywords: Return autocorrelation; Mispricing; Factor enhancement; Conditional asset pricing; Mean-variance optimization.

*Nottingham University Business School China, University of Nottingham Ningbo China. Email: yiyan.qian@nottingham.edu.cn.

[†]Nottingham University Business School China, University of Nottingham Ningbo China. Email: ying.jiang@nottingham.edu.cn.

[‡]Corresponding author. Nottingham University Business School China, University of Nottingham Ningbo China. Email: xiaoquan.liu@nottingham.edu.cn.

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1 Introduction

The recent literature in equity markets documents the momentum in factor portfolios: the long-short characteristic portfolios that outperform in the past continue to generate higher returns in the future ([Arnott et al., 2023](#); [Ehsani and Linnainmaa, 2022a](#); [Gupta and Kelly, 2019](#)). Such phenomenon is referred to as the *factor momentum*, which captures information different from the traditional momentum in the cross section and time series of stock returns ([Jegadeesh and Titman, 1993](#); [Moskowitz et al., 2012](#)). [Ehsani and Linnainmaa \(2022a\)](#) analyze the time-series factor momentum and argue that factor returns are positively autocorrelated. The factor momentum is rationalized by mispricing as arbitrageurs shy away from trading against persistent sentiment ([Arnott et al., 2023](#); [Ehsani and Linnainmaa, 2022a](#); [Kozak et al., 2018](#)).

Our paper extends the idea of time-series factor momentum to commodity futures markets. We examine the existence of the factor momentum in these markets and explore possible economic explanations. Following [Ehsani and Linnainmaa \(2022b\)](#), we evaluate the efficiency of commodity pricing models based on the factor momentum and improve the performance of these models with enhanced factors. By exploring the factor momentum, our paper aims to advance our understanding of asset pricing and risk premium in commodity futures markets, which are of vital importance to investors and firms with a commercial interest in these markets.

The commodity futures markets have become increasingly important over the past few decades, where the trading volume and investment have increased sharply ([Basak and Pavlova, 2016](#); [Cheng and Xiong, 2014](#); [Tang and Xiong, 2012](#)). According to the World Federation of Exchanges, more than 10 billion commodity futures contracts have

been traded in 2021. A growing number of mimicking strategies based on cross-sectional characteristics are shown to generate sizable profits in these markets, including the basis ([Gorton and Rouwenhorst, 2006](#); [Kojen et al., 2018](#)), basis momentum ([Boons and Prado, 2019](#)), and hedging pressure ([Kang et al., 2020](#); [Szymanowska et al., 2014](#)). In addition, commodity pricing models built upon these mimicking strategies are developed in the literature with empirical success ([Bakshi et al., 2019](#); [Boons and Prado, 2019](#); [Yang, 2013](#)).

Why could factor momentum exist in commodity futures markets? The literature in equity markets rationalizes the factor momentum via a mispricing channel ([Ehsani and Linnainmaa, 2022a](#)). Theoretical and empirical evidence suggests that the mispricing, which matches factor risks, cannot be traded away because arbitrageurs avoid taking factor risks so the mispricing remains and leads to momentum in factor returns. This mechanism thus relates the factor momentum to the slow-moving mispricing correction due to the limits of arbitrage ([Ma et al., 2023](#)). In the literature on commodity futures markets, there is also evidence of mispricing and limited arbitrage ([Da et al., 2023](#); [Fernandez-Perez et al., 2018](#); [Girma and Paulson, 1999](#)). Hence, the factor momentum could exist in commodity futures markets as a result of mispricing.

In our baseline tests, we use a sample of 36 commodity futures contracts traded in the US and UK from January 1985 to May 2022 to construct ten well-documented factors. We uncover strong factor momentum over the one-month horizon which leads to sizable economic gains: a commodity factor earns an average monthly return of 66 basis points (bps) (t -statistic = 3.57) following a month of positive returns, and an insignificant return of almost zero following a month of negative returns. Furthermore, a factor momentum strategy that buys (sells) the factors with positive (negative) returns in the

previous month earns an annualized return of 4.37% (t -statistic = 2.80) and significant risk-adjusted returns under popular pricing models of [Boons and Prado \(2019\)](#), [Bakshi et al. \(2019\)](#), and [Yang \(2013\)](#). These risk-adjusted returns are driven by the factor autocorrelation based on the return decomposition of [Leippold and Yang \(2021\)](#).

We next examine whether the factor momentum stems from the mispricing. We follow [Ehsani and Linnainmaa \(2022a\)](#) and conduct the following test: if the mispricing channel is valid, the factor momentum should concentrate on systematic factors, i.e., mimicking portfolios that explain systematic variations in returns. Our results are consistent with this prediction. We extract the principal components (PCs) from commodity factors in our sample and observe that the factor momentum predominantly concentrates on high-eigenvalue PCs, which serve as proxies for systematic factors. This concentration is pronounced in the whole sample period and in the subsample after the financialization of commodities.

The evidence of autocorrelated factor returns underlying the factor momentum suggests a *mismatch* between factor's returns and volatility if the factor volatility does not change with past factor returns ([Ehsani and Linnainmaa, 2022b](#)). This mismatch gives rise to mean-variance inefficient factor portfolios and sends distorted signals to market participants ([Grinblatt and Titman, 1987](#)). Hence, we empirically examine the efficiency of existing commodity pricing models. Based on [Boons and Prado \(2019\)](#), we find that the commodity market factor and basis momentum factor are mean-variance inefficient: while the factor's returns increase with past returns, the factor's volatility remains largely unchanged.¹ So we construct the time-series efficient factors, which take dynamic positions in the original factor to minimize the variance while maintaining expected returns

¹ We always include the pricing models of [Yang \(2013\)](#) and [Bakshi et al. \(2019\)](#) alongside [Boons and Prado \(2019\)](#) throughout the study. The three models yield qualitatively the same results.

(Ehsani and Linnainmaa, 2022b). We find that these efficient factors exhibit higher Sharpe ratios. For instance, the Sharpe ratio improvements in efficient market and basis momentum factors are 0.24 (t -statistic = 2.71) and 0.13 (t -statistic = 2.76), respectively. More importantly, the pricing model with efficient factors is more informative than its standard version at the 1% significance level (p -value = 0.00) in the spanning tests. As an example, the alphas of efficient market factor (t -statistic = 2.63) and efficient basis momentum (t -statistic = 2.67) are both significant at the 1% level after controlling for the standard model. These findings are robust after the financialization of commodity futures and clearly indicate that the time-series transformation substantially improves the pricing power of commodity factor models and provides economic benefits for mean-variance utility investors.

Finally, we generalize our baseline results on the factor momentum to international markets. We focus on a larger dataset of developed markets with contracts traded in the US, Canada, UK, France, and Singapore, and 68 contracts in developing markets of Brazil, China, India, and Russia. The results based on these developed and developing markets are consistent with the baseline findings in that 1) the factor momentum exists, and 2) the performance of commodity pricing models can be improved with mean-variance efficient commodity factors. The international evidence lends strong support for the baseline results.

Our study adds to the literature in two directions. First, we extend the literature on factor momentum by providing novel empirical evidence from commodity futures markets. The existing literature documents the factor momentum in equity (Ehsani and Linnainmaa, 2022a; Arnott et al., 2023; Gupta and Kelly, 2019; Ma et al., 2023; Yan and Yu, 2023), foreign exchange (Zhang, 2022), and options (Kaefer et al., 2023) markets.

Leippold and Yang (2021) question the existence of factor momentum and argue that the profitability of factor momentum strategy in equity markets is a manifestation of mean factor returns instead of factor return predictability. Our empirical analysis in commodity futures markets provides an out-of-sample investigation beyond the equity markets, contributing to the debate on the existence and rationale of the factor momentum.

We also contribute to the strand of literature on conditional asset pricing. These studies largely focus on the equity markets, which enhance the equity pricing models by utilizing *conditioning* information, such as the covariance structure (Daniel and Titman, 1997; Daniel et al., 2020), lagged factor volatility (Moreira and Muir, 2017; Zimmermann, 2022), lagged factor returns (Ehsani and Linnainmaa, 2022b), aggregated signals (Hollstein and Prokopczuk, 2023), and conditional risk (Gormsen and Jensen, 2022). Existing studies in commodity futures markets are limited to discussing the implication of the lagged factor volatility for commodity factor improvement (Sakkas and Tessaromatis, 2020; Kang and Kwon, 2021). We instead improve commodity pricing models based on lagged factor returns and our results shed light on the importance of conditional asset pricing in commodity futures markets.

The rest of this paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the data and commodity factors returns. In Section 4, we examine the existence of factor momentum and explore its economic rationale. Section 5 conducts the time-series transformation. In Section 6, we explore the factor momentum in international commodity futures markets. Finally, Section 7 concludes.

2 Literature review

Factor momentum

A collection of research explores the factor momentum at the time-series level. Based on 22 equity factors, [Ehsani and Linnainmaa \(2022a\)](#) document the time-series factor momentum by showing that equity factor returns are positively autocorrelated. They rationalize this phenomenon by mispricing and analyze the relation between factor momentum and individual stock momentum in [Jegadeesh and Titman \(1993\)](#). [Gupta and Kelly \(2019\)](#) extend [Ehsani and Linnainmaa \(2022a\)](#) and confirm the existence of factor momentum in Asia-Pacific and European equity markets. Across different asset classes, [Zhang \(2022\)](#) documents significant factor momentum in currency markets, whereas [Kaefer et al. \(2023\)](#) show that the factor momentum exists in options markets.

Meanwhile, [Leippold and Yang \(2021\)](#) question the existence of time-series factor momentum and argue that this phenomenon is merely a byproduct of factor returns. By decomposing the factor momentum into the buy-and-hold and pure factor timing portfolio components, they find that the factor momentum returns based on 210 US equity factors are driven by the mechanical exposure to the factor premium, while the predictability of past performance is empirically weak. There are also studies investigating the factor momentum in the cross section and suggest that factors outperforming others in the past continue to outperform in the future ([Arnott et al., 2023](#); [Avramov et al., 2017](#); [Lewellen, 2002](#); [McLean and Pontiff, 2016](#)).

Conditional asset pricing and factor enhancement

The recent literature points out that traditional asset pricing models tend to be inefficient because factor-mimicking portfolios are not mean-variance efficient ([Daniel et al., 2020](#); [Ehsani and Linnainmaa, 2022b](#); [Gormsen and Jensen, 2022](#); [Hollstein and Prokopczuk, 2023](#); [Zimmermann, 2022](#)). To mitigate this problem, these studies resort to conditional asset pricing, which utilizes conditioning information to improve the mean-

variance efficiency of factor portfolios. Through factor enhancement, the factor portfolio's returns become a better representation of the factor risk. Moreover, pricing models based on enhanced factor portfolios would move closer to the efficient frontier and achieve improved asset pricing power.

In particular, [Daniel et al. \(2020\)](#) study the mean-variance inefficiency of factors in the Fama-French five-factor model and construct characteristic-efficient portfolios. [Barroso and Santa-Clara \(2015\)](#), [Daniel and Moskowitz \(2016\)](#), and [Moreira and Muir \(2017\)](#) improve the mean-variance inefficient equity factors by introducing volatility-managed portfolios. [Zimmermann \(2022\)](#) takes a step further by applying this volatility management to enhance the pricing power of equity factor models. [Ehsani and Linnainmaa \(2022a\)](#) utilize the conditioning information in factor momentum to construct time-series efficient factors. [Hollstein and Prokopczuk \(2023\)](#) and [Gormsen and Jensen \(2022\)](#) enhance the market factor based on aggregated signals and conditional risk, respectively.

In the commodity futures markets, [Sakkas and Tessaromatis \(2020\)](#) construct volatility-managed factor portfolios and show that the managed momentum yields a significantly higher Sharpe ratio than the standard commodity momentum. Meanwhile, [Kang and Kwon \(2021\)](#) eliminate the impact of look-ahead bias and exploit the out-of-sample volatility management for real-time investors. However, no significant improvement in momentum, commodity market factor, basis momentum, and two components of the basis momentum has been identified.

In our study, we focus on the commodity futures markets and examine the factor momentum and time-series factor enhancement in this important asset class.

3 Data and factor returns

We collect settlement prices, open interests, and trading volume of 36 actively traded commodity futures products from January 1985 to May 2022, including 29 contracts traded in the US and 7 in the UK, from the Bloomberg.² Our dataset covers commodities in four sectors, i.e., agriculture (16), energy (6), livestock (3), and metal (10). Table A.1 summarizes detailed information for these commodity futures contracts.

Following Gorton et al. (2013) and Han and Kong (2022), we use the first-nearby contracts to calculate the commodity futures excess returns on a fully collateralized futures position based on the roll-over strategy as follows:

$$Return_{i,d+1}^{(1)} = \begin{cases} \frac{F_{i,d+1}^{(1)}}{F_{i,d}^{(1)}} - 1, & \text{if no contract is rolled over at day } d, \\ \frac{F_{i,d+1}^{(1)}}{F_{i,d}^{(2)}} - 1, & \text{otherwise,} \end{cases} \quad (1)$$

where $F_{i,d}^{(1)}$ and $F_{i,d+1}^{(1)}$ denote the settlement price of the first nearby futures contract i on day d and $d + 1$, respectively, and $F_{i,d}^{(2)}$ is the settlement price of the second nearby futures contract on day d . We compound daily excess returns to obtain monthly returns for each commodity.

Table 1 summarizes the descriptive statistics for commodity futures excess returns. The mean returns vary across commodities. The highest annualized mean returns come from tin, gasoline blendstock, and soybean meal. Tin generates the highest annualized Sharpe ratio at 0.38. There is a large variation in the volatility of commodity futures returns. The annualized standard deviation reaches the lowest in live cattle (14.29%)

² Our sample starts from January 1985 with 7 factors. Due to the data availability, the liquidity, hedging pressure, and open interest factors begin in September 1989, January 1994, and March 1995, respectively.

and the highest in natural gas (46.29%). More than half of the contracts are positively skewed and all contracts show leptokurtic returns.

We construct ten commodity factors based on these contracts. Each factor is related to an existing trading strategy in the literature. In particular, the basis captures the information in the slope of futures term structure ([Gorton and Rouwenhorst, 2006](#); [Koijen et al., 2018](#)); momentum exploits the cross-sectional predictability of past performance ([Erb and Harvey, 2006](#); [Miffre and Rallis, 2007](#); [Shen et al., 2007](#)); basis momentum relates to the slope and curvature of the term structure of futures returns ([Boons and Prado, 2019](#)); hedging pressure reflects the mismatch in the amount of hedging and speculating activity ([Kang et al., 2020](#); [Szymanowska et al., 2014](#)); skewness relates to the mispricing from investor with cumulative prospect theory preferences and selective hedging practices ([Fernandez-Perez et al., 2018](#)); open interest reflects future price initiation ([Hong and Yogo, 2012](#); [Szymanowska et al., 2014](#)); currency β captures the exposure to the changes in the U.S. dollar versus a basket of foreign currencies ([Erb and Harvey, 2006](#); [Szymanowska et al., 2014](#)); inflation β reflects the impact from unexpected inflation to commodity futures markets ([Szymanowska et al., 2014](#)); and liquidity captures the liquidity risk of commodity futures trading ([Szymanowska et al., 2014](#)). The detailed description of these commodity factors is summarized in the [Appendix](#).

We construct factor mimicking portfolios by sorting commodities into quintiles in month $t - 1$ according to the predictive characteristic of a trading strategy. The long-short portfolios involve buying (selling) the quintile with the highest (lowest) predicted returns in month t with an equal-weighting scheme. All portfolios are rebalanced monthly. We use the S&P Goldman Sachs Commodity Index (GSCI) to represent the commodity market (MKT).³

³ The S&P GSCI is the industry standard benchmark index for commodity investment and composed

Table 2 summarizes the descriptive statistics for commodity factors. The skewness factor shows the strongest performance among all with a highly significant annualized return of 10.33%. The skewness factor is associated with mispricing in commodity futures markets as investors prefer to overprice positively-skewed commodities and underprice negatively-skewed commodities (Fernandez-Perez et al., 2018). The substantial returns of the skewness factor suggest potentially sizable mispricing in commodity futures markets, which is consistent with the economic rationale behind the factor momentum. The basis, basis momentum, and momentum factors generate large excess returns of 7.27%, 8.19%, and 8.21%, respectively, and are highly significant. The notable performance of these three factors underpins their popularity in the existing commodity pricing models. The profitability of other strategies, i.e., currency β , inflation β , liquidity, hedging pressure, and open interest, is weaker. Overall, our results for the profitability of these commodity factors are largely consistent with evidence in the literature (see Boons and Prado, 2019; Daskalaki et al., 2014; Fernandez-Perez et al., 2018, for instance).

4 Factor momentum

4.1 Autocorrelated factors

We test the predictability of a factor's past returns for its future returns over 5 different horizons: 1, 3, 6, 9, and 12 months. We estimate two predictive time-series regressions of contracts in the agricultural, energy, livestock, and metals sectors. We also use an alternative commodity factor as the equally weighted portfolio of all commodity futures in our sample with qualitatively the same results. These results are available upon request.

as follows:

$$Return_{f,t} = \alpha + \beta \times PastReturn_{f,t-h} + \epsilon_{f,t}, \quad (2)$$

$$Return_{f,t} = \alpha + \beta \times PastSign_{f,t-h} + \epsilon_{f,t}, \quad (3)$$

where $Return_{f,t}$ is factor f 's returns in month t , and $PastReturn_{f,t-h}$ is the factor f 's average returns over the past h -month horizon prior to month t . The $PastSign_{f,t-h}$ is a dummy variable that is equal to one if factor f 's average returns over the past h -month horizon before month t are positive, and zero otherwise. These regressions are estimated for each factor and pooled across all factors. Note that the momentum factor is excluded from the pooled regression and the factor momentum construction to prevent the mechanical correlation between factor momentum and individual momentum ([Ehsani and Linnainmaa, 2022a](#)).

[Table 3](#) summarizes regression results and our main interest lies in pooled tests, which evaluate the overall level of factor autocorrelation in commodity futures markets. Panel A shows that the pooled predictability of prior-month factor returns for future returns is statistically significant across commodity factors. Based on nine non-momentum factors, an average factor's monthly returns significantly increase with its returns in the previous month at the 1% level (t -statistic = 4.76). Following a month of positive returns, this average factor earns 66 bps (t -statistic = 3.57) per month. On the contrary, when the factor's prior month returns are negative, the monthly returns are insignificant and close to zero. For individual factors, all of them exhibit positive slope coefficients in both regressions suggesting prevalent positive factor autocorrelation. Meanwhile, although commodity factors' unconditional returns are positive in [Table 2](#) except for the liquidity

factor, results in Table 3 show that six out of nine factors earn negative returns after a month of negative returns, i.e., in the return-on-sign regression, which also suggests autocorrelated factor returns.

For longer horizons, the predictability of factor premium becomes slightly weaker. In Panel B, the pooled results for the three-month horizon show that monthly returns of an average factor positively relate to its return over the past three months at the 5% level (t -statistic = 2.19) in the return-on-return regression. Besides, for the return-on-sign specification over the three-month horizon, the average factor earns 13 bps (t -statistic = 0.96) after a negative return over the past three months but returns increase to 45 bps (t -statistic = 2.43) after a positive return. For individual factors, several individual factors exhibit negative slope coefficients, implying that the existence of factor momentum is mixed. Results are largely insignificant over six-, nine-, and 12-month horizons reported in Table A.2.

In summary, the commodity futures factors are positively autocorrelated on average and this autocorrelation is the most significant over the one-month horizon. It represents appealing opportunities to commodity investors who believe in the factor momentum: they could potentially make investment decisions based on the sign of a factors' unconditional returns in the previous month.

4.2 Profitability of factor momentum strategy

We exploit the substantial pooled factor autocorrelation by buying factors with positive past returns and selling those with negative returns in a long-short trading strategy. Results in Table 4 show that the factor momentum strategy is most pronounced over the one-month horizon. In Panel A, we observe that the strategy earns an annualized

return of 4.37% (t -statistic = 2.80) and delivers significant risk-adjusted returns under the pricing models of Boons and Prado (2019), Yang (2013), and Bakshi et al. (2019) at 4.54% (t -statistic = 2.87), 4.56% (t -statistic = 2.88), and 3.83% (t -statistic = 2.54), respectively.

Moreover, we scrutinize returns to the winner, loser, and benchmark portfolios: the winner (loser) portfolios contain factors with positive (negative) past returns, and the benchmark portfolio takes equal weight across all factors. We expect the winner (loser) portfolio to outperform (underperform) the benchmark if the factors' past performance predicts their future returns. We observe that in Panel A, the winner portfolio generates a highly significant annualized return of 8.17% (t -statistic = 4.40) and outperforms the benchmark portfolio, which earns 4.43% (t -statistic = 3.52). Meanwhile, the loser portfolio underperforms the benchmark portfolio with an insignificant annualized return of -0.40% (t -statistic = -0.19). In Figure 1, we display the cumulative returns of the three portfolios over our sample period.

The profitability of the momentum strategy over the three-month horizon is weaker. Panel B shows that the three-month factor momentum generates an annualized return of 3.11% (t -statistic = 2.20) with a risk-adjusted return of 2.58% (t -statistic = 1.82) under the Boons and Prado (2019) model. Furthermore, the annualized returns of the winner portfolio decline to 6.85% (t -statistic = 3.58), while returns of the loser portfolio increase to 1.52% (t -statistic = 0.76). The winners (losers) still outperform (underperform) the benchmark indicating that factor returns over the past three months are still predictive of future factor returns. We provide results for longer horizons over six, nine, and 12 months in Table A.3 and find that the factor momentum strategy fails to generate significant profits.

To sum up, the factor momentum strategy based on the short-term factor autocorrelation yields substantial profits, which are the most sizable over the one-month horizon. These findings highlight the significant predictive power of factor returns over the short horizon.

4.3 Factor momentum decomposition

According to [Leippold and Yang \(2021\)](#), returns of the factor momentum stem from two sources: the mechanical exposure to factor risk premium in a static way, the predictable factor timing ability, or a combination of both. We decompose the factor momentum into two portfolios accordingly: the first portfolio is based on a static buy-and-hold strategy (BH) without active timing and earns the factor premium. It involves taking long (short) position in factors in month t with positive (negative) average returns up to and include month $t - 1$. The second portfolio is based on a pure factor timing strategy (FT) to capture the economic benefits of factor return predictability, i.e., it takes long (short) position in factors in month t when their average monthly returns over the past horizons are higher (lower) than their mean returns up and include month $t - 1$. The strategy only works when a factor's past returns deviate from the prevailing mean returns thus ruling out the factor premium.

In [Table 5](#), we summarize the decomposition results over the one- and three-month horizons in Panels A and B, respectively. In Panel A, we observe that the FT strategy generates a highly significant return of 4.23% (t -statistic = 2.80) per year, while the BH strategy earns an insignificant return of 1.41% (t -statistic = 0.96). These suggest that the timing ability of the one-month factor autocorrelation is strong in commodity futures markets, whereas the commodity factor premium is weak on average. It also indicates

that the significant profitability of the one-month factor momentum is not driven by the mean factor premium.

Next, we perform spanning tests to examine the relation between factor momentum and two component portfolios and find out whether the two components help explain the factor momentum strategy profits. We also conduct these spanning tests in the opposite way to regress BH and FT portfolio returns on the factor momentum returns. The spanning tests yield consistent results. First, while both BH and FT components partially explain factor momentum returns, the explanatory power of the FT portfolio is much stronger than that of the BH strategy. For instance, in Panel A Columns (1) and (2), the slope coefficient of the FT portfolio is highly significant and sizable at 0.99% (t -statistic = 71.85), while that of the BH portfolio is merely 0.11% (t -statistic = 2.22).

Results in Columns (1) and (2) also show that the FT can fully explain the factor momentum, but BH cannot. This is evidenced by the intercepts and adjusted R^2 . We note that the factor momentum generates a significant alpha of 0.35% (t -statistic = 2.70) after controlling for the BH strategy, and the adjusted- R^2 is only 1.1%. On the contrary, the factor momentum produces an insignificant alpha of almost zero after controlling for the FT strategy with an adjusted R^2 of 92.0%. Finally, it is not surprising to find that the factor momentum portfolio dominates both BH and FT components. The reverse spanning tests in Columns (3) and (4) generate insignificant intercepts in both regressions indicating that the factor momentum fully explains BH and FT strategies.

In Panel B, results for the three-month factor momentum are qualitatively the same. We find the FT strategy yields significant returns of 2.50% (t -statistic = 1.74) per year while the BH strategy generates an insignificant return of 1.37% (t -statistic = 0.94). Furthermore, the FT portfolio shows superior explanatory power for the factor momentum

with a slope coefficient of 0.92% (t -statistic = 55.24). [Table A.4](#) reports the decomposition results for the factor momentum over the past six, nine, and 12 months.

In summary, the decomposition results suggest that the commodity factor momentum profits are driven by the factors' active timing ability rather than a byproduct of the commodity factor premium. These differ from the findings in [Leippold and Yang \(2021\)](#) in equity markets and highlight the economic significance of factor autocorrelation in commodity futures markets.

4.4 Factor momentum and mispricing

We rationalize the existence of factor momentum by exploring the mispricing explanation ([Ehsani and Linnainmaa, 2022a](#)). We extract nine PCs from nine non-momentum factors and consider the high- (low-) eigenvalue PCs as systematic (non-systematic) risks. In month t , we utilize the information up to month $t - 1$ to calculate PC factors' returns, and construct the factor momentum portfolio returns. This procedure is a fully out-of-sample approach to avoid the look-ahead bias. We use daily factor returns starting in January 1985 to compute the eigenvectors and require at least 5 years of data to extract PCs. The returns of factor momentum strategies therefore begin in February 1990.

Based on the prediction of the mispricing channel, the strongest momentum is expected to manifest in high-eigenvalue PCs, i.e., systematic factors. Results in [Table 6](#) are consistent with this prediction. We find that the strategy trading high PC factors generates monthly returns of 61 bps (t -statistic = 3.14) and offers significant risk-adjusted returns at the 1% level (t -statistic = 3.16). The factor momentum in the low PC factors is weaker with returns of 44 bps (t -statistic = 2.40).

We also conduct the spanning tests to evaluate whether the high eigenvalue factors

generate stronger momentum than low eigenvalue factors. Specifically, Panel B spans the momentum in the high PC factors ($FMOM_t^{HighPC}$) by the momentum in the low PC factors ($FMOM_t^{LowPC}$) based on the following regression:

$$FMOM_t^{HighPC} = \alpha + \beta \times FMOM_t^{LowPC} + X'_t + \varepsilon_t, \quad (4)$$

where α represents incremental returns of $FMOM_t^{HighPC}$ over $FMOM_t^{LowPC}$. The X'_t denotes a vector of the MKT and basis momentum factors. Panel C conducts the reverse spanning tests, which regress the factor momentum in low PC factors by momentum in high PC factors.

Our main focus is on the intercept in Panels B and C. Panel B shows that the momentum in high PC factors generates significant abnormal returns over that in low PC factors with a significant α of 0.41. These indicate that the factor momentum in high eigenvalue factors significantly outperforms that in low eigenvalue factors. Moreover, results in Panel C show that the spanning tests do not work in the opposite way, as the momentum in low PC factors is fully explained by that in high PC factors with insignificant α . Our findings point to the concentration of factor momentum in the high PC set, i.e., systematic risks, and are in line with the mispricing explanation.

For robustness, we conduct a subsample analysis after the financialization in commodity futures markets. The financialization of commodities, which begins in 2004, is associated with a rapid boom of commodity investment and an unprecedented entry of institutional investors into the markets (Cheng and Xiong, 2014; Singleton, 2014; Tang and Xiong, 2012). This transition is widely accepted as a structural break in commodity futures markets (Basak and Pavlova, 2016; Da et al., 2023; Goldstein and Yang, 2022).

We show that our results are qualitatively the same after the financialization. In the last

columns of Panel A, the high PC factors still generate a stronger factor momentum than the low PC set in the subsample with a monthly return of 53 bps (t -statistic = 2.23) and significant risk-adjusted returns (t -statistic = 2.45), while the strategies based on low PC set exhibit insignificant returns. And the momentum in high PC factors earns significant abnormal returns (t -statistic = 2.14) in the spanning tests after controlling for the momentum in the low PC set.

In summary, we find that the factor momentum concentrates on important systematic factors, consistent with the prediction of the mispricing channel. This is pronounced in the full sample and after the financialization of commodities.

5 Time-series efficient factors and pricing models

If factor returns are positively autocorrelated while a factor's return variance does not change with its past returns, there exists a disconnection between factor premium and factor risk. This disconnection indicates that the factor portfolios are not mean-variance efficient (MVE) and violate the fundamental assumption for asset pricing models, making the models inefficient (Grinblatt and Titman, 1987). In this section, we focus on the commodity factor model of Boons and Prado (2019) for our baseline analysis - results based on Yang (2013) and Bakshi et al. (2019) are qualitatively the same.

5.1 Mean-variance inefficient factors

To examine the mean-variance efficiency of each factor, we regress the factor returns and return volatility, respectively, on factor returns in the previous month. Table 7 Panel A shows that the returns of both factors are positively autocorrelated. For example, the prior-month returns of MKT significantly predict its future returns at 16 bps (t -statistic

= 3.43). Panel B examines the predictive power of prior-month factor returns for future factor volatility. We find that the slope coefficients of lagged factor returns on factor return volatility are -1 bps (t -statistic = -1.15) for the MKT and -3 bps (t -statistic = -1.74) for the basis momentum. These results indicate that the factor risk does not increase with lagged factor returns, different from the positive time-series dependency between factor returns.

This pricing inefficiency remains pronounced after the financialization. Specifically, the factor returns are significantly autocorrelated for both MKT (t -statistic = 3.17) and basis momentum (t -statistic = 2.45), whereas the relation between lagged factor return and factor return volatility is insignificant for both factors. Taken together, we observe that while factor returns significantly increase with past returns, factor volatilities are largely insensitive to past returns. This mismatch suggests that the factor portfolio may load *unpriced* sources of risk making them inefficient in the mean-variance framework.

5.2 Time-series efficient factors

To construct time-series efficient factors, we allow investors to leverage forecasting signals and dynamically update their position between inefficient factors and risk-free assets to minimize the variance for a given expected return (Ehsani and Linnainmaa, 2022b; Ferson and Siegel, 2001). As investors take levered or conservative positions in the factor, this dynamic weighting scheme produces MVE portfolios that offer the same returns at the lowest volatility. Specifically, an efficient factor invests x in the original factor portfolio and the remainder in the risk-free asset as follows:

$$r_{e,t} = x_t \times r_t, \quad (5)$$

where $r_{e,t}$ and r_t are the month t excess returns of the efficient factor and original factor, respectively, and x_t is the optimal weight on the original factor in month t .

Ehsani and Linnainmaa (2022b) narrow down the forecasting signal to be the prior-month factor's returns and engage in the time-series transformation. Hence, for each factor, the optimal weight x_t in month t is expressed as follows:

$$x_t = \mu \times \frac{SR^2 + 1}{SR^2 + \rho^2} \times \frac{\mu(1 - \rho) + \rho r_{t-1}}{(\mu(1 - \rho) + \rho r_{t-1})^2 + \sigma_\epsilon^2}, \quad (6)$$

where μ , SR , and ρ denote the mean returns, Sharpe ratio, and autocorrelation coefficient of the original factor, respectively. The variance of the error term $\sigma_\epsilon^2 = (1 - \rho^2) \sigma^2$, where σ is the standard deviation of factor returns. To construct the efficient factors in month t , we estimate factors' means, volatility, and autocorrelations based on information up to month $t - 1$ to compute optimal weight for month $t - 1$. This approach is again fully out-of-sample without the look-ahead bias.

Table 8 Panel A reports annualized Sharpe ratios of the original and efficient factors. We note that the time-series transformation pushes up the Sharpe ratio of the MKT from 0.29 to 0.52 and basis momentum from 0.43 to 0.56 in the model of Boons and Prado (2019). We then follow Daniel et al. (2020), Moreira and Muir (2017), and Zimmermann (2022) to use the Jensen's alpha to determine whether the differences in Sharpe ratios are statistically significant. Results in Panel B show that the improvement is significant at the 1% level for both the MKT and basis momentum: the efficient MKT generates a monthly alpha of 0.42 (t -statistic = 2.71) and the efficient basis momentum has an alpha of 25 bps (t -statistic = 2.76) over the whole sample period. We observe significantly improved Sharpe ratios for the subsample after the financialization, as the efficient MKT and basis momentum enhance the SR by 0.40 and 0.38, respectively, and significant at

the 1% level.

To summarize, the time-series transformation based on the [Boons and Prado \(2019\)](#) model exhibits superior MVE in comparison to the original factors with substantially larger Sharpe ratios. The gains are economically and statistically significant for both the MKT and basis momentum factors over the whole sample and after the financialization of commodities.

5.3 Improvement in asset pricing models

We compare the pricing performance of the [Boons and Prado \(2019\)](#) model based on efficient MKT and basis momentum factors with the performance based on standard MKT and basis momentum via spanning tests ([Ehsani and Linnainmaa, 2022b](#); [Zimmermann, 2022](#)). In particular, we regress the efficient factor by the standard factor model as follows:

$$Factor_t^e = \alpha + \beta_1 \times MKT_t + \beta_2 \times Basismom_t + \epsilon_t, \quad (7)$$

where MKT_t and $Basismom_t$ are the returns of standard MKT and basis momentum factors, respectively, and $Factor_t^e$ denotes returns of the efficient MKT or efficient basis momentum, all in month t . We then conduct regressions in the opposite way of each standard factor against the efficient factor model as follows:

$$Factor_t = \alpha + \beta_1 \times MKT_t^e + \beta_2 \times Basismom_t^e + \epsilon_t, \quad (8)$$

where MKT_t^e and $Basismom_t^e$ are the returns in month t of efficient MKT and basis momentum, respectively. Our main focus is on the intercept term, which measures the incremental information of each factor against the right-hand-side model.

In [Table 9](#) Panel A for the whole sample period, we find that the efficient MKT exhibits a significant alpha against the standard factor model at 42 bps (t -statistics = 2.63) per month, and efficient basis momentum offers a significant alpha at 24 bps (t -statistics = 2.67). In the [Gibbons et al. \(1989\)](#) (GRS) test, the null hypothesis of joint zero intercepts for the efficient MKT and basis momentum factors is rejected at the 1% level (p -value = 0.00) suggesting strong incremental information in the efficient factors against a standard version of the factor model. In the reverse regressions in Panel B, we find both standard factors are uninformative in the efficient factor model with insignificant intercepts individually and jointly. Moreover, the enhancement of the factor model remains substantial after the financialization in the subsample analysis.

The above results suggest that the time-series transformation significantly enhances the pricing power of the [Boons and Prado \(2019\)](#) model. This is achieved by adopting mean-variance efficient common commodity factors based on the conditional information in the factor momentum leading to an improved match between portfolio returns and volatility. Qualitatively the same improvement is achieved for the pricing models of [Yang \(2013\)](#) and [Bakshi et al. \(2019\)](#) in [Table A.5](#).

6 International evidence

To generalize our findings of the factor momentum in commodity futures markets, we use a larger sample of contracts in developed and developing markets. While our main tests focus on 36 commodity futures traded in major commodity exchanges in the US and UK, we include four more contracts traded in Canada, France, and Singapore from 1985 to 2022. Detailed information on the data is summarized in [Table A.6](#).

This comprehensive set of data based on developed markets provides consistent evi-

dence for the factor momentum. [Table 10](#) shows that a factor's past one-month returns significantly predict its future returns by 56 bps (t -statistic = 3.31), and the factor momentum strategy of buying (selling) factors with positive (negative) past returns earns a significant return of 3.57% (t -statistic = 3.74). Furthermore, [Table A.7](#) shows that high-eigenvalue factors generate stronger factor momentum than low-eigenvalue factors, which is consistent with the mispricing channel. And the time-series transformation still significantly improves the pricing power of the [Boons and Prado \(2019\)](#) model in [Table A.8](#).

Our final piece of jigsaw comes from evidence based on developing markets. We collect data on 68 actively traded commodity futures contracts in Brazil, China, India, and Russia from the Bloomberg and the China Stock Market & Accounting Research (CSMAR) database. For developing markets, we use 2004 as the start year to have at least eight contracts for constructing commodity factors in the cross section. Detailed information on these contracts and summary statistics of commodity futures excess returns are summarized in [Table A.9](#) and [Table A.10](#).

In [Table 11](#), we observe significant predictability of a factor's prior-month returns on its future returns in developing commodity markets: an average factor earns a monthly return of 90 bps (t -statistic = 4.44) following a month of positive returns and an insignificant negative return following a month of negative returns. Meanwhile, the factor momentum strategy generates significant returns of 5.65% (t -statistic = 3.74) and significant risk-adjusted returns. We also observe in [Table A.11](#) that the commodity factor momentum concentrates more on high-eigenvalue factors. Finally, we exploit the factor momentum to construct efficient factors in [Table A.12](#) and find that the efficient version of the [Boons and Prado \(2019\)](#) model is more informative against its original version at

the 1% significance level (p -value = 0.00) in GRS tests. Hence, the results for factor momentum in developing commodity futures markets are highly consistent with the baseline results and the evidence in developed commodity futures markets.

7 Conclusion

This paper explores the time-series factor momentum in global commodity futures markets. We find that the factor momentum in commodity futures markets is statistically significant and at its strongest over the one-month horizon. It leads to sizable economic gains for investors in a factor momentum strategy. We rationalize this short-term phenomenon based on the mispricing and arbitrage activity: the mispricing aligning with factors premium cannot be corrected because arbitrageurs avoid taking factor risks. This explanation is supported empirically as we find that the factor momentum concentrates more on high-eigenvalue factors.

The existence of factor momentum poses a challenge for popular pricing models as autocorrelated factor returns suggest mean-variance inefficient pricing models. To improve the efficiency of individual common factors, we construct the time-series efficient factors by exploiting the factor momentum. We show that the factor models based on efficient factors significantly outperform the original factor models. We further provide international evidence that these findings are prevalent across developed and developing markets.

Our paper reveals the time-varying risk premium of commodity factors and shows that the pricing efficiency of existing benchmark models can be improved by timing factors' past returns. These findings help us gain a better understanding of factor-based investment and offer important implications to practitioners and researchers.

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Figure 1: Cumulative returns of factor momentum strategy

This figure displays cumulative returns of the factor momentum strategy, which involves buying factors with positive past returns (FMOM winners) and selling those with negative past returns (FMOM losers) over the past one (Panel A) and three (Panel B) months. The benchmark portfolios (EqStrategy) take equal weight of all factors.

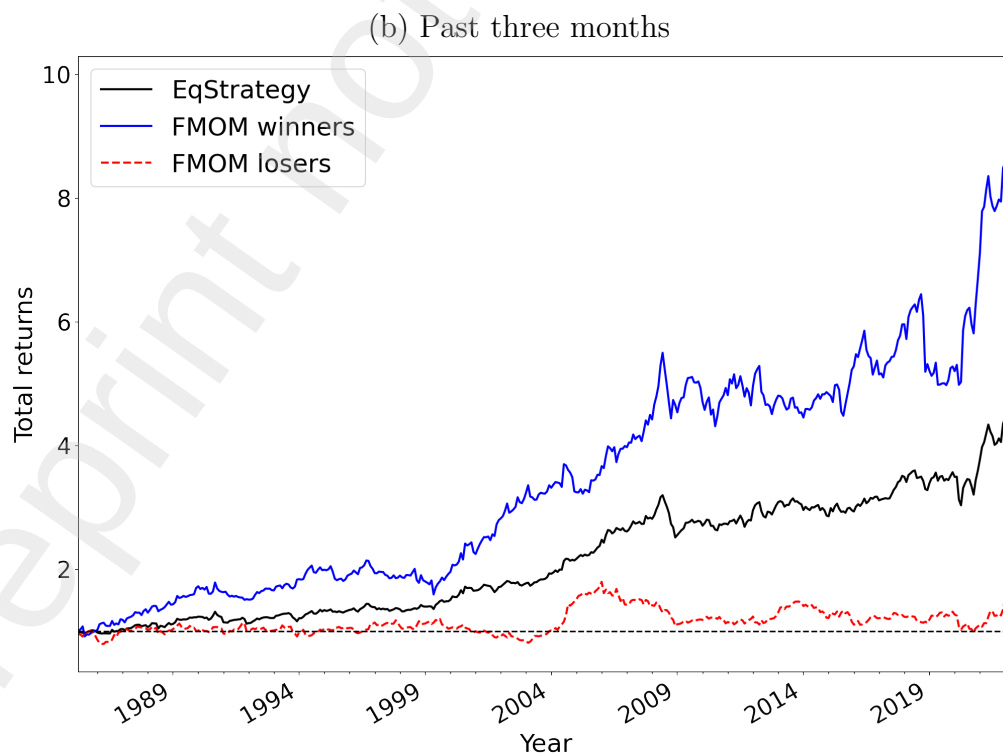
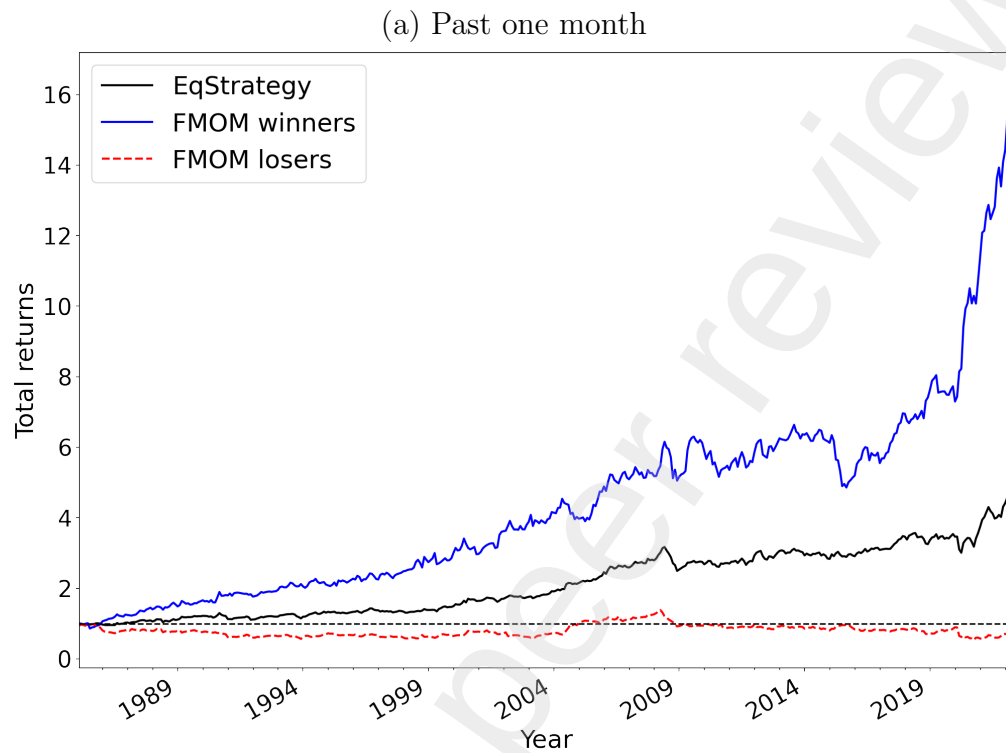


Table 1: Summary statistics of commodity futures returns

This table reports the mean (in percent), standard deviation (SD, in percent), Sharpe ratio (SR), skewness (Skew), kurtosis (Kurt), maximum (Max, in percent), and minimum (Min, in percent) of excess futures returns of 36 commodities. The sample period is from January 1985 to May 2022.

Category	Contract	Mean	SD	SR	Skew	Kurt	Max	Min
Agriculture	Butter	-1.23	25.96	-0.05	0.18	3.30	32.81	-27.79
	Cocoa	-6.75	27.79	-0.24	0.12	5.60	29.69	-28.78
	Coffee	-8.52	35.36	-0.24	0.53	3.70	41.54	-36.92
	Corn	-5.62	25.68	-0.22	0.28	4.88	38.52	-25.88
	Kansas Wheat	-1.60	25.82	-0.06	0.13	7.46	30.74	-27.22
	Lumber	-0.59	35.36	-0.02	0.14	6.20	43.39	-46.68
	Milk	2.34	28.18	0.08	0.30	4.46	40.46	-32.58
	Oats	-1.81	31.63	-0.06	0.99	8.75	66.21	-31.12
	Orange Juice	-3.45	30.70	-0.11	0.31	8.95	44.93	-30.16
	Rough rice	-6.77	25.00	-0.27	0.24	5.12	38.52	-25.90
	Soybean Meal	7.50	24.90	0.30	0.11	4.79	26.35	-24.92
	Soybean oil	-1.95	24.48	-0.08	-0.13	4.86	23.57	-29.04
	Soybeans	3.16	22.42	0.14	-0.31	21.21	21.25	-24.95
	Sugar	0.89	32.14	0.03	0.21	6.19	42.31	-35.11
	Wheat	-5.36	26.32	-0.20	0.04	4.35	32.02	-28.30
	Cotton	-0.76	25.51	-0.03	-0.09	5.36	24.48	-25.67
Energy	Crude oil, Brent	6.60	33.76	0.20	-0.82	4.66	42.11	-63.38
	Crude oil, WTI	5.03	36.12	0.14	-0.61	4.12	48.31	-68.01
	Gasoil	6.09	32.71	0.19	-0.35	5.06	33.54	-40.49
	Gasoline, Unleaded	4.07	40.05	0.10	-2.78	5.67	28.89	-89.51
	Gasoline, blendstock	7.85	40.20	0.20	0.36	6.25	42.35	-38.83
	Heating Oil	6.25	31.74	0.20	-0.18	3.62	31.45	-38.37
	Natural Gas	-15.38	46.29	-0.33	0.06	3.27	42.07	-39.02
Livestock	Feeder Cattle	0.95	14.60	0.07	-0.31	8.54	12.90	-17.47
	Lean Hogs	-4.82	26.77	-0.18	-0.40	7.18	25.55	-30.06
	Live Cattle	1.84	14.29	0.13	-0.47	5.30	14.87	-23.69
Metals	Aluminum	-2.35	19.29	-0.12	0.01	6.91	14.84	-18.61
	Copper	3.79	25.08	0.15	-0.50	5.40	29.23	-45.37
	Gold	1.29	15.17	0.08	0.03	9.51	14.75	-20.41
	Lead	3.30	27.87	0.12	-0.33	9.01	23.58	-32.05
	Nickel	6.36	34.39	0.18	0.02	5.79	32.33	-27.76
	Palladium	7.49	31.60	0.24	-0.27	7.67	38.52	-41.45
	Platinum	1.65	22.02	0.08	-0.55	4.91	29.14	-37.93
	Silver	-0.82	27.54	-0.03	0.02	3.96	26.18	-32.82
	Tin	8.95	23.55	0.38	0.06	4.90	23.70	-24.10
	Zinc	0.43	26.00	0.02	-0.49	6.01	24.41	-42.37

Table 2: Commodity factor portfolios

This table reports the start of sample period, mean (in percent) and its t -statistics, standard deviation (SD, in percent), and Sharpe ratio (SR) for commodity factor portfolios. The sample period ends in May 2022 for all contracts.

Factor	Start	Mean	t -stat	SD	SR
MKT	198501	5.90	1.74	20.67	0.29
Basis	198501	7.27	2.96	15.05	0.48
Momentum	198501	8.21	2.20	22.88	0.36
Basis momentum	198501	8.19	2.63	19.04	0.43
Skewness	198501	10.33	3.28	19.24	0.54
Currency β	198501	4.03	1.04	23.69	0.17
Inflation β	198501	1.34	0.34	23.81	0.06
Liquidity	198909	-0.91	-0.30	17.43	-0.05
Hedging pressure	199401	1.30	0.41	17.10	0.08
Open interest	199503	0.83	0.27	16.30	0.05

Table 3: Factors returns conditional on their own past returns

This table reports estimation results for two time-series regressions for individual factors and pooled across all factors conditional on returns over the past one (Panel A) and three (Panel B) months. In the *return-to-return* specification, a factor's returns are regressed on the moving average of its past returns, whereas in the *return-to-sign* specification, a factor's returns are regressed on a dummy variable indicating positive moving average in the past. The sample period is from January 1985 to May 2022. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A. Past one month

	Returns				Sign of returns			
	α	$t(\alpha)$	β	$t(\beta)$	α	$t(\alpha)$	β	$t(\beta)$
<i>Pooled</i>	0.35***	3.78	0.08***	4.76	0.02	0.16	0.66***	3.57
MKT	0.42	1.49	0.16***	3.43	-0.04	-0.09	0.96*	1.69
Basis	0.55***	2.65	0.09**	1.98	0.09	0.30	0.93**	2.25
Momentum	0.66**	2.10	0.03	0.70	0.67	1.43	0.02	0.02
Basis momentum	0.63**	2.41	0.08	1.62	0.49	1.25	0.33	0.63
Skewness	0.84***	3.15	0.03	0.64	0.55	1.35	0.53	0.99
Currency β	0.32	0.99	0.06	1.18	-0.11	-0.23	0.81	1.24
Inflation β	0.09	0.28	0.06	1.24	-0.26	-0.57	0.71	1.09
Liquidity	-0.07	-0.26	0.07	1.42	-0.23	-0.66	0.34	0.67
Hedging pressure	0.11	0.41	0.06	1.05	-0.04	-0.11	0.31	0.58
Open interest	0.07	0.26	0.08	1.36	-0.23	-0.63	0.59	1.14

Panel B. Past three months

	Returns				Sign of returns			
	α	$t(\alpha)$	β	$t(\beta)$	α	$t(\alpha)$	β	$t(\beta)$
<i>Pooled</i>	0.36***	3.85	0.06**	2.19	0.13	0.96	0.45**	2.43
MKT	0.43	1.50	0.14*	1.81	-0.09	-0.20	0.99*	1.73
Basis	0.54**	2.57	0.14*	1.74	0.26	0.83	0.65	1.57
Momentum	0.67**	2.11	0.01	0.14	0.79*	1.65	-0.20	-0.32
Basis momentum	0.59**	2.23	0.11	1.36	0.38	0.93	0.46	0.86
Skewness	0.78***	2.86	0.10	1.20	0.71*	1.66	0.23	0.42
Currency β	0.38	1.17	-0.07	-0.88	0.47	0.96	-0.20	-0.30
Inflation β	0.11	0.35	-0.01	-0.16	0.03	0.06	0.18	0.27
Liquidity	-0.06	-0.23	0.03	0.41	-0.12	-0.35	0.13	0.26
Hedging pressure	0.07	0.26	0.17**	1.92	-0.41	-1.04	0.95*	1.76
Open interest	0.08	0.32	-0.01	-0.15	-0.07	-0.19	0.30	0.57

Table 4: Profitability of factor momentum strategy

This table summarizes returns of the factor momentum portfolios (FMOM) involving buying factors with positive past returns (Winner) and selling those with negative past returns (Loser) in the past one (Panel A) and three (Panel B) months. The benchmark portfolios (EqStrategy) take equal weight of all factors. The α and $t(\alpha)$ are risk-adjusted returns and their t -statistics based on pricing models of Boons and Prado (2019) (BP2), Yang (2013) (Y2), and Bakshi et al. (2019) (BGR3). The portfolio mean returns (in percent) and their t -statistics, standard deviation (SD, in percent), and Sharpe ratio (SR) are all annualized. The sample period is from January 1985 to May 2022. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	Mean	t -stat	SD	SR	α_{BP2}	$t(\alpha_{BP2})$	α_{Y2}	$t(\alpha_{Y2})$	α_{BGR3}	$t(\alpha_{BGR3})$
<i>Panel A. Past one month</i>										
FMOM	4.37***	2.80	9.55	0.46	4.54***	2.87	4.56***	2.88	3.83**	2.45
Winner	8.17***	4.40	11.35	0.72	6.31***	3.62	6.28***	3.58	5.65***	3.25
Loser	-0.40	-0.19	13.12	-0.03	-2.53	-1.24	-2.48	-1.20	-1.74	-0.85
EqStrategy	4.43***	3.52	7.68	0.58	2.10**	2.28	2.00**	2.17	1.97**	2.12
<i>Panel B. Past three months</i>										
FMOM	3.11**	2.20	8.61	0.36	2.58*	1.82	2.77*	1.95	1.75	1.29
Winner	6.85***	3.58	11.67	0.59	4.36***	2.61	4.25**	2.53	3.29**	2.01
Loser	1.52	0.76	12.09	0.13	-0.15	-0.08	-0.19	-0.10	0.68	0.36
EqStrategy	4.47***	3.55	7.68	0.58	2.17**	2.36	2.02**	2.18	1.98**	2.12

Table 5: Factor momentum decomposition

This table decomposes factor momentum portfolios into the buy-and-hold (BH) and factor timing (FT) components over the past one (Panel A) and three (Panel B) months. We report the mean (in percent) and its t -statistics, standard deviation (SD, in percent), and Sharpe ratio (SR) for the BH and FT, all annualized. We also report the spanning test results and the t -statistics are in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

<i>Panel A. Past one month</i>				
	Summary statistics			
	Mean	t -stat	SD	SR
BH	1.41	0.96	8.94	0.16
FT	4.23***	2.80	9.23	0.46
	Spanning tests			
	(1)	(2)	(3)	(4)
	FMOM	FMOM	BH	FT
BH	0.11** (2.22)			
FT		0.99*** (71.85)		
FMOM			0.10** (2.22)	0.93*** (71.85)
Constant	0.35*** (2.70)	0.01 (0.39)	0.08 (0.67)	0.01 (0.41)
Obs.	448	448	448	448
Adj. R^2	1.1%	92.0%	1.1%	92.0%
<i>Panel B. Past three months</i>				
	Summary statistics			
	Mean	t -stat	SD	SR
BH	1.37	0.94	8.94	0.15
FT	2.50*	1.74	8.76	0.29
	Spanning tests			
	(1)	(2)	(3)	(4)
	FMOM	FMOM	BH	FT
BH	0.26*** (6.00)			
FT		0.92*** (55.24)		
FMOM			0.28*** (6.00)	0.95*** (55.24)
Constant	0.23** (2.02)	0.07 (1.61)	0.04 (0.35)	-0.04 (-0.89)
Obs.	446	446	446	446
Adj. R^2	7.5%	87.3%	7.5%	87.3%

Table 6: Factor momentum in high- and low-eigenvalue factors

This table summarizes the factor momentum performance across different sets of PC factors. The high- (low-) eigenvalue set includes PC factors with high (low) eigenvalues. Panel A reports monthly descriptive statistics for momentum strategies, including the mean returns (in percent) and their t -statistics. Panels B and C report spanning test results for the full sample and subsample after the financialization. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

<i>Panel A. Factor momentum in subsets of PC factors ordered by eigenvalues</i>						
Sets of PCs	Full sample			Subsample		
	Mean	t -stat	$t(\alpha_{BP2})$	Mean	t -stat	$t(\alpha_{BP2})$
High	0.61***	3.14	3.16	0.53**	2.23	2.45
Low	0.44**	2.40	2.34	0.24	1.15	1.22

<i>Panel B. Factor momentum and high-eigenvalue PC factors</i>		
	FMOM ^{HighPC}	
	Full sample	Subsample
α	0.41** (2.33)	0.48** (2.14)
FMOM ^{LowPC}	0.49*** (10.08)	0.43*** (6.04)
Controls	Y	Y
N	388	221
Adj. R^2	20.5%	13.8%

<i>Panel C. Factor momentum and low-eigenvalue PC factors</i>		
	FMOM ^{LowPC}	
	Full sample	Subsample
α	0.16 (1.00)	0.06 (0.31)
FMOM ^{HighPC}	0.43*** (10.08)	0.33*** 6.04
Controls	Y	Y
N	388	221
Adj. R^2	20.3%	14.5%

Table 7: Predictability of the MKT and basis momentum factors

This table examines the mean-variance efficiency of the MKT and basis momentum in the Boons and Prado (2019) model. We perform univariate regressions to examine the predictability of factor's returns for its future returns and volatility and report the slope coefficients and their corresponding t -statistics for the full sample and subsample after the financialization. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

<i>Panel A. Predicting returns</i>				
	Full sample		Subsample	
	β_{return}	$t(\beta_{return})$	β_{return}	$t(\beta_{return})$
MKT	0.16***	3.43	0.21***	3.17
Basis momentum	0.08	1.62	0.16**	2.45

<i>Panel B. Predicting volatility</i>				
	Full sample		Subsample	
	β_{return}	$t(\beta_{return})$	β_{return}	$t(\beta_{return})$
MKT	-0.01	-1.15	-0.02	-1.23
Basis momentum	-0.03*	-1.74	-0.05	-1.46

Table 8: Time-series efficient factors

Panel A compares the Sharpe ratios of standard factors (SR) with those of corresponding time-series efficient factors (SR*) for the MKT and basis momentum in the Boons and Prado (2019) model, and Panel B regresses each time-series efficient factor on its standard version, for both the full and subsample after the financialization of commodities. The t -statistics are in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

<i>Panel A. Improvement in SR</i>				
	Full sample		Subsample	
	MKT	Basis momentum	MKT	Basis momentum
SR	0.29	0.43	0.38	0.31
SR*	0.52	0.56	0.78	0.70

<i>Panel B. Univariate regressions</i>				
	Full sample		Subsample	
	MKT*	Basis momentum*	MKT*	Basis momentum*
α	0.42*** (2.71)	0.25*** (2.76)	0.64*** (2.91)	0.51*** (2.83)
MKT	0.21*** (8.03)		0.22*** (6.90)	
Basis momentum		0.85*** (51.96)		0.47*** (13.80)
Adj. R^2	12%	85%	20%	47%

Table 9: Time-series efficient asset pricing models

This table compares the performance of the Boons and Prado (2019) model based on the time-series efficient factors (MKT* and Basis momentum*) and standard factors (MKT and Basis momentum) for the full and subsample. Panel A spans the efficient model by the standard model, whereas Panel B does the opposite. We report the alphas and their t -statistics in parentheses, and the Gibbons et al. (1989) (GRS) test F -statistics and their p -values in square brackets. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

<i>Panel A. Efficient model regressed on standard model</i>		
	Full sample	Subsample
	α_{BP2}	α_{BP2}
MKT*	0.41*** (2.63)	0.64*** (2.91)
Basis momentum*	0.24*** (2.67)	0.50*** (2.72)
GRS F -statistic	6.40*** [0.00]	8.15*** [0.00]
<i>Panel B. Standard model regressed on efficient model</i>		
	Full sample	Subsample
	α_{BP2}^*	α_{BP2}^*
MKT	0.17 (0.63)	-0.02 (-0.05)
Basis momentum	-0.13 (-1.34)	-0.26 (-0.94)
GRS F -statistic	1.08 [0.34]	0.45 [0.64]

Table 10: Factor momentum across more developed markets

Panel A reports estimation results for two time-series regressions for individual factors and pooled across all factors conditional on returns over the past one month. In the *return-to-return* specification, a factor's returns are regressed on the moving average of its past returns, whereas in the *return-to-sign* specification, a factor's returns are regressed on a dummy variable indicating positive moving average in the past. Panel B summarizes returns of the factor momentum portfolios (FMOM) involving buying factors with positive past returns (Winner) and selling those with negative past returns (Loser) in the past month, and the benchmark portfolios (EqStrategy) take equal weight of all factors. These tests are based on a larger set of developed markets. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A. Factors returns conditional on own past returns

	Returns				Sign of returns			
	α	$t(\alpha)$	β	$t(\beta)$	α	$t(\alpha)$	β	$t(\beta)$
Pooled	0.29***	3.46	0.07***	4.26	0.01	0.12	0.56***	3.31
MKT	0.51**	2.55	0.09**	2.00	0.39	1.31	0.32	0.80
Basis	0.70***	2.78	0.07	1.56	0.66*	1.71	0.16	0.32
Momentum	0.92***	2.99	0.00	-0.01	0.71	1.53	0.36	0.58
Basis momentum	0.02	0.12	0.14***	2.98	-0.18	-0.78	0.41	1.24
Skewness	0.11	0.41	0.06	1.05	-0.04	-0.11	0.31	0.58
Currency β	0.37	1.23	0.06	1.35	0.13	0.29	0.48	0.79
Inflation β	0.13	0.43	0.06	1.24	-0.40	-0.92	1.08*	1.76
Liquidity	-0.23	-1.01	0.04	0.69	-0.37	-1.17	0.28	0.60
Hedging pressure	0.04	0.18	0.08	1.48	-0.59*	-1.71	1.26***	2.59
Open interest	0.80***	3.14	0.04	0.76	0.58	1.46	0.42	0.81

Panel B. Profitability of factor momentum strategy

	Mean	t -stat	SD	SR	α_{BP2}	$t(\alpha_{BP2})$	α_{Y2}	$t(\alpha_{Y2})$	α_{BGR3}	$t(\alpha_{BGR3})$
FMOM	5.65***	3.74	6.46	0.87	0.37***	2.85	0.49***	3.80	0.39***	3.04
Winner	7.14***	3.31	9.23	0.77	0.38**	2.47	0.44***	2.87	0.41***	2.62
Loser	-2.22	-0.94	9.98	-0.22	-0.22	-1.18	-0.32*	-1.78	-0.18	-1.02
EqStrategy	2.41	1.41	7.32	0.33	0.02	0.20	0.03	0.37	0.07	0.77

Table 11: Factor momentum across developing markets

Panel A reports estimation results for two time-series regressions for individual factors and pooled across all factors conditional on returns over the past one month. In the *return-to-return* specification, a factor's returns are regressed on the moving average of its past returns, whereas in the *return-to-sign* specification, a factor's returns are regressed on a dummy variable indicating positive moving average in the past. Panel B summarizes returns of the factor momentum portfolios (FMOM) involving buying factors with positive past returns (Winner) and selling those with negative past returns (Loser) in the past month, and the benchmark portfolios (EqStrategy) take equal weight of all factors. These tests are based on developing markets. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A. Factors returns conditional on own past returns

	Returns				Sign of returns			
	α	$t(\alpha)$	β	$t(\beta)$	α	$t(\alpha)$	β	$t(\beta)$
<i>Pooled</i>	0.18*	1.83	0.12***	5.46	-0.28*	-1.86	0.90***	4.44
MKT	0.06	0.26	0.19***	2.83	-0.68*	-1.88	1.43***	2.89
Basis	0.95***	2.78	0.04	0.57	0.79	1.43	0.30	0.44
Momentum	1.45***	3.70	-0.01	-0.13	0.83	1.32	0.98	1.24
Basis momentum	1.02***	3.80	0.04	0.63	0.54	1.23	0.81	1.50
Skewness	0.06	0.21	0.14**	2.09	-0.29	-0.72	0.69	1.23
Currency β	0.14	0.37	0.14**	2.01	-0.24	-0.41	0.75	0.96
Inflation β	0.02	0.04	0.31***	4.44	-1.41**	-2.50	2.61***	3.39
Liquidity	-0.16	-0.49	0.12*	1.84	-0.42	-0.95	0.50	0.78
Hedging pressure	-0.24	-0.84	0.01	0.20	-0.53	-1.33	0.59	1.03
Open interest	-0.10	-0.44	-0.12*	-1.72	0.02	0.07	-0.24	-0.52

Panel B. Profitability of factor momentum strategy

	Mean	t -stat	SD	SR	α_{BP2}	$t(\alpha_{BP2})$	α_{Y2}	$t(\alpha_{Y2})$	α_{BGR3}	$t(\alpha_{BGR3})$
FMOM	5.65***	3.74	6.46	0.87	0.37***	2.85	0.49***	3.80	0.39***	3.04
Winner	7.14***	3.31	9.23	0.77	0.38**	2.47	0.44***	2.87	0.41***	2.62
Loser	-2.22	-0.94	9.98	-0.22	-0.22	-1.18	-0.32*	-1.78	-0.18	-1.02
EqStrategy	2.41	1.41	7.32	0.33	0.02	0.20	0.03	0.37	0.07	0.77

Appendix: Description of commodity factors

Basis: The basis represents the price difference between futures and spot commodity contracts. For each commodity j , we follow Boons and Prado (2019) to use the following equation to calculate basis in month t :

$$\text{Basis}_t^j = \frac{F_{j,t}^{(2)}}{F_{j,t}^{(1)}} - 1, \quad (9)$$

where $F_{j,t}^{(2)}$ and $F_{j,t}^{(1)}$ are respectively the commodity j 's futures prices for the second and first nearby contracts.

Momentum: For each commodity j , we follow Boons and Prado (2019) to calculate the characteristic for momentum in month t as the cumulative excess returns during past 12-month period:

$$\text{Momentum}_t^j = \prod_{s=t-11}^t \left(1 + R_{j,t,s}^{(1)}\right) - 1, \quad (10)$$

where $R_{j,t}^{(1)}$ represents the commodity j 's returns in month t for the first nearby futures contracts.

Basis momentum: For each commodity j , we follow Boons and Prado (2019) to quantify the characteristics for basis momentum in month t based on the following equation:

$$\text{BasisMomentum}_t^j = \prod_{s=t-11}^t \left(1 + R_{j,t,s}^{(1)}\right) - \prod_{s=t-11}^t \left(1 + R_{j,t,s}^{(2)}\right), \quad (11)$$

where $R_{j,t,s}^{(2)}$ and $R_{j,t,s}^{(1)}$ represent commodity j 's returns for the second and first nearby futures contract.

Liquidity: We follow Szymanowska et al. (2014) to calculate the liquidity as the averaged Amihud et al. (1997)'s measure, which divides the volume on a trading day by

the absolute return on that trading day, over the two most recent months.

Skewness: We calculate the skewness following [Fernandez-Perez et al. \(2018\)](#). For each commodity j , the skewness in month t is defined as follows:

$$Sk_t^j = \frac{\frac{1}{D} \sum_{d=1}^D \left(R_{j,d}^{(1)} - \mu_{j,t} \right)^3}{\sigma_{j,t}^3}, \quad (12)$$

where $R_{j,d}^{(1)}$ denotes the commodity j 's daily futures returns for the first-nearby contracts at day d during the past 12 months. D , μ , and σ respectively represent daily observation number, mean, and standard deviation estimates for daily returns over the past 12-month period.

Currency β : Following [Szymanowska et al. \(2014\)](#), the currency β is constructed from a 60-month rolling regression of monthly first-nearby commodity futures returns on changes in the U.S. dollar versus a basket of foreign currencies.

Commodity market: We calculate the monthly returns of S&P GSCI to indicate the commodity futures markets. For the analysis based on international data, we construct a portfolio that takes equal weight in all contracts in the sample to indicate the market.

Inflation β : Following [Szymanowska et al. \(2014\)](#), the inflation β factor is constructed according to the slope coefficient based on the prior 60-month regressions of monthly first-nearby commodity futures returns on unexpected inflation, which measured as the change in one-month CPI inflation rate.

Hedging pressure: We follow [Bakshi et al. \(2019\)](#) to calculate the hedging pressure for commodity j in month t as the net short open interest over the total open interest of

commercial traders during the last 12 months of the commodity t :

$$HP_t^j = \sum_{i=0}^{12-1} \frac{Short_{j,t-i} - Long_{j,t-i}}{Long_{j,t-i} + Short_{j,t-i}} \quad (13)$$

where $Short_{j,t-i}$ and $Long_{j,t-i}$ respectively denote the short and long positions of commercial traders.

Open Interest: For each commodity j , we follow [Sakkas and Tessaromatis \(2020\)](#) to quantify the sorting variable in month t based on following equation:

$$\Delta OI_{j,t} = OI_{j,t} - OI_{j,t-1}, \quad (14)$$

where $OI_{j,t}$ and $OI_{j,t-1}$ stand for commodity j 's aggregated open interest in month t and $t - 1$, respectively.

Table A.1: Commodity futures data

This table summarizes 36 commodity futures contracts in the baseline analysis, including their sector, futures exchange, market of trading, the Bloomberg ticker, and date of the first observation in the sample.

Category	Contract	Exchange	Market	Ticker	Start
Agriculture	Butter	CME	U.S.	V6	200509
	Cocoa	ICE	U.S.	CC	198001
	Coffee	ICE	U.S.	KC	198001
	Corn	CBOT	U.S.	C	198001
	Cotton	ICE	U.S.	CT	198001
	Kansas Wheat	CBOT	U.S.	KW	198001
	Lumber	CME	U.S.	LB	198604
	Milk	CME	U.S.	DA	199601
	Oats	CBOT	U.S.	O	198001
	Orange Juice	ICE	U.S.	JO	198001
	Rough Rice	CBOT	U.S.	RR	198812
	Soybean	CBOT	U.S.	S	198001
	Soybean Meal	CBOT	U.S.	SM	198001
	Soybean Oil	CBOT	U.S.	BO	198001
	Sugar	ICE	U.S.	SB	198001
	Wheat	CBOT	U.S.	W	198001
Energy	Crude Oil Brent	ICE	U.K.	CO	198806
	Crude Oil WTI	NYMEX	U.S.	CL	198303
	Gasoil	ICE	U.K.	QS	198907
	Gasoline blendstock	NYMEX	U.S.	HU	200510
	Gasoline Unleaded	NYMEX	U.S.	XB	198604
	Heating Oil	NYMEX	U.S.	HO	198607
	Natural Gas	NYMEX	U.S.	NG	199004
Livestock	Feeder Cattle	CME	U.S.	FC	198001
	Lean Hogs	CME	U.S.	LH	198604
	Live Cattle	CME	U.S.	LC	198001
Metals	Aluminum	LME	U.K.	LA	199707
	Copper	NYMEX	U.S.	HG	198812
	Gold	NYMEX	U.S.	GC	198001
	Lead	LME	U.K.	LL	199707
	Nickel	LME	U.K.	LN	199707
	Palladium	NYMEX	U.S.	PA	198604
	Platinum	NYMEX	U.S.	PL	198604
	Silver	NYMEX	U.S.	SI	198001
	Tin	LME	U.K.	LT	199707
	Zinc	LME	U.K.	LX	199707

Table A.2: Factors returns conditional on their own past returns over the past six, nine, and 12 months

This table reports estimation results for two time-series regressions for individual factors and pooled across all factors conditional on returns over the past six (Panel A), nine (Panel B), and 12 (Panel C) months. In the *return-to-return* specification, a factor's returns are regressed on the moving average of its past returns, whereas in the *return-to-sign* specification, a factor's returns are regressed on a dummy variable indicating positive moving average in the past. The sample period is from January 1985 to May 2022. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A. Past six months

	Returns				Sign of returns			
	Intercept		Slope		Intercept		Slope	
	α	$t(\alpha)$	β	$t(\beta)$	α	$t(\alpha)$	β	$t(\beta)$
<i>Pooled</i>	0.38***	4.02	0.00	0.07	0.39***	2.76	-0.02	-0.11
Market	0.46	1.60	0.12	1.14	0.38	0.86	0.22	0.38
Basis	0.57***	2.63	0.12	1.10	0.34	0.97	0.48	1.11
Momentum	0.69**	2.13	-0.06	-0.48	0.75	1.55	-0.16	-0.26
Basis momentum	0.65**	2.39	-0.03	-0.23	0.93**	2.08	-0.46	-0.83
Skewness	0.88***	3.13	-0.04	-0.36	0.86*	1.91	-0.03	-0.05
Currency β	0.42	1.26	-0.07	-0.57	0.66	1.35	-0.49	-0.74
Inflation β	0.13	0.41	-0.09	-0.71	0.22	0.48	-0.19	-0.29
Liquidity	-0.09	-0.34	-0.05	-0.41	-0.08	-0.22	-0.02	-0.03
Hedging pressure	0.07	0.27	0.04	0.30	0.02	0.05	0.11	0.21
Open interest	0.05	0.19	-0.13	-0.94	0.37	1.02	-0.69	-1.31

Panel B. Past nine months

	Returns				Sign of returns			
	Intercept		Slope		Intercept		Slope	
	α	$t(\alpha)$	β	$t(\beta)$	α	$t(\alpha)$	β	$t(\beta)$
<i>Pooled</i>	0.39***	4.07	-0.03	-0.54	0.42***	2.91	-0.08	-0.41
Market	0.51*	1.73	0.00	0.03	0.34	0.74	0.26	0.45
Basis	0.53**	2.37	0.19	1.46	-0.01	-0.01	0.98**	2.24
Momentum	0.67**	2.06	0.03	0.22	0.58	1.22	0.21	0.33
Basis momentum	0.70**	2.53	-0.07	-0.50	0.95**	1.97	-0.42	-0.73
Skewness	0.88***	3.06	0.03	0.24	0.86*	1.83	0.07	0.13
Currency β	0.44	1.32	-0.27*	-1.76	1.19**	2.41	-1.49**	-2.27
Inflation β	0.10	0.30	-0.11	-0.69	0.34	0.73	-0.50	-0.75
Liquidity	-0.12	-0.46	-0.04	-0.26	0.20	0.57	-0.69	-1.34
Hedging pressure	0.08	0.31	0.02	0.13	-0.07	-0.19	0.32	0.58
Open interest	0.05	0.20	-0.12	-0.69	0.09	0.23	-0.08	-0.15

Panel C. Past 12 months

	Returns				Sign of returns			
	Intercept		Slope		Intercept		Slope	
	α	$t(\alpha)$	β	$t(\beta)$	α	$t(\alpha)$	β	$t(\beta)$
<i>Pooled</i>	0.39***	4.07	-0.03	-0.54	0.42***	2.91	-0.08	-0.41
Market	0.51*	1.73	0.00	0.03	0.34	0.74	0.26	0.45
Basis	0.56**	2.43	0.20	1.33	0.63*	1.69	0.07	0.15
Momentum	0.69**	2.06	0.00	-0.02	0.59	1.18	0.16	0.25
Basis momentum	0.71**	2.54	-0.04	-0.22	1.24**	2.51	-0.76	-1.31
Skewness	0.92***	3.08	0.00	0.01	0.96**	1.99	-0.06	-0.11
Currency β	0.38	1.13	-0.16	-0.86	0.79	1.59	-0.82	-1.24
Inflation β	0.05	0.14	-0.03	-0.15	-0.03	-0.06	0.15	0.23
Liquidity	-0.06	-0.22	0.21	1.23	-0.41	-1.19	0.76	1.48
Hedging pressure	0.05	0.19	0.00	0.02	0.02	0.06	0.06	0.10
Open interest	0.04	0.14	-0.15	-0.73	-0.07	-0.18	0.19	0.35

Table A.3: Profitability of factor momentum strategy over the past six, nine, and 12 months

This table summarizes returns of the factor momentum portfolios (FMOM) involving buying factors with positive past returns (Winner) and selling those with negative past returns (Loser) in the past six (Panel A), nine (Panel B), and 12 (Panel C) months. The benchmark portfolios (EqStrategy) take equal weight of all factors. The α and $t(\alpha)$ are risk-adjusted returns and their t -statistics based on pricing models of Boons and Prado (2019) (BP2). The mean portfolio returns (in percent) and their t -statistics, standard deviation (SD, in percent), and Sharpe ratio (SR) are annualized. The sample period is from January 1985 to May 2022. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Strategy	Mean	t -stat	SD	SR	α_{BP2}	$t(\alpha_{BP2})$
<i>Panel A. Past six months</i>						
FMOM	0.16	0.11	9.02	0.02	-0.63	-0.429
Winner	4.47**	2.30	11.80	0.38	1.80	1.055
Loser	3.50*	1.76	12.04	0.29	2.04	1.059
EqStrategy	4.48***	3.53	7.70	0.58	2.14**	2.311
<i>Panel B. Past nine months</i>						
FMOM	0.38	0.26	8.78	0.04	-0.89	-0.63
Winner	4.96***	2.60	11.58	0.43	2.03	1.238
Loser	3.57*	1.76	12.30	0.29	2.70	1.351
EqStrategy	4.41***	3.45	7.74	0.57	3.41*	1.835
<i>Panel C. Past 12 months</i>						
FMOM	1.70	1.19	8.62	0.20	0.16	0.116
Winner	5.19***	2.76	11.34	0.46	2.05	1.32
Loser	1.86	0.90	12.51	0.15	0.90	0.441
EqStrategy	4.47***	3.47	7.78	0.57	3.86**	2.118

Table A.4: Factor momentum decomposition over the past six, nine, and 12 months

This table decomposes factor momentum portfolios into the buy-and-hold (BH) and factor timing (FT) components over the past six (Panel A), nine (Panel B), and 12 (Panel C) months. We report the mean (in percent) and its t -statistics, standard deviation (SD, in percent), and Sharpe ratio (SR) for the BH and FT, all annualized. We also report the spanning test results and the t -statistics are in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

<i>Panel A. Past six months</i>				
	Summary statistics			
	Mean	t -stat	SD	SR
BH	1.18	0.80	8.96	0.13
FT	0.00	0.00	8.87	0.00
	Spanning tests			
	(1)	(2)	(3)	(4)
	FMOM	FMOM	BH	FT
BH	0.39*** (8.81)			
FT		0.92*** (45.51)		
FMOM			0.38*** (8.81)	0.89*** (45.51)
Constant	-0.02 (-0.26)	0.01 (0.26)	0.09 (0.82)	-0.01 (-0.24)
Obs.	443	443	443	443
Adj. R^2	1.5%	82.4%	1.5%	82.4%
<i>Panel B. Past nine months</i>				
	Summary statistics			
	Mean	t -stat	SD	SR
BH	1.43	0.98	8.85	0.16
FT	-0.84	-0.61	8.40	-0.10
	Spanning tests			
	(1)	(2)	(3)	(4)
	FMOM	FMOM	BH	FT
BH	0.43*** (9.98)			
FT		0.92*** (38.67)		
FMOM			0.43*** (9.98)	0.84*** (38.67)
Constant	-0.02 (-0.17)	0.10* (1.68)	0.11 (0.96)	-0.10* (-1.77)
Obs.	440	440	440	440
Adj. R^2	18.5%	77.3%	18.5%	77.3%
<i>Panel C. Past 12 months</i>				
	Summary statistics			
	Mean	t -stat	SD	SR
BH	1.39	0.94	8.96	0.16
FT	1.36	0.97	8.44	0.16
	Spanning tests			
	(1)	(2)	(3)	(4)
	FMOM	FMOM	BH	FT
BH	0.48*** (11.84)			
FT		0.84*** (30.68)		
FMOM			0.51*** (11.84)	0.81*** (30.68)
Constant	0.09 (0.84)	0.05 (0.69)	0.04 (0.40)	0.00 (-0.02)
Obs.	437	437	437	437
Adj. R^2	24.4%	68.4%	24.4%	68.4%

Table A.5: Time-series efficient models of Yang (2013) and Bakshi et al. (2019)

This table compares the performance of the Yang (2013) (Panel A) and Bakshi et al. (2019) (Panel B) models based on the time-series efficient factors (MKT*, Basis*, and Momentum*) and standard factors (MKT, Basis, and Momentum) for the full and subsample. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. See also notes to Table 9.

<i>Panel A. Time-series efficient model of Yang (2013)</i>		
	<i>Full sample</i>	<i>Subsample</i>
	α_{Yang2}	α_{Yang2}
MKT*	0.42*** (2.71)	0.64*** (2.89)
Basis*	0.26** (2.16)	0.38*** (2.58)
GRS <i>F</i> -statistic	5.72*** [0.00]	6.88*** [0.00]
	α_{Yang2}^*	α_{Yang2}^*
MKT	0.15 (0.55)	-0.13 (-0.31)
Basis	0.09 (0.59)	-0.11 (-0.56)
GRS <i>F</i> -statistic	0.32 [0.73]	0.19 [0.83]
<i>Panel B. Time-series efficient model of Bakshi et al. (2019)</i>		
	<i>Full sample</i>	<i>Subsample</i>
	α_{BGR3}	α_{BGR3}
MKT*	0.36** (2.32)	0.61*** (2.85)
Basis*	0.24* (1.99)	0.37** (2.54)
Momentum*	0.13 (1.12)	-0.05 -0.26
GRS <i>F</i> -statistic	3.19** [0.02]	4.47*** [0.00]
	α_{BGR3}^*	α_{BGR3}^*
MKT	0.15 (0.56)	-0.14 (-0.33)
Basis	0.12 (0.79)	-0.11 (-0.55)
Momentum	-0.01 (-0.12)	-0.07 (-0.22)
GRS <i>F</i> -statistic	0.31 [0.82]	0.17 [0.92]

Table A.6: Additional contracts in developed markets

This table summarizes information for four additional commodity futures contracts traded in developed markets and their summary statistics. See also notes to [Table 1](#) and [Table A.1](#).

Panel A. Contract information

Category	Contract	Exchange	Market	Ticker	Start
Agriculture	Canola	ICE	Canada	RS	198201
	Rapeseed	MATIF	France	IJ	199901
	Rubber	SGX	Singapore	OR	199707
Metals	Iron Ore	SGX	Singapore	SCO	201304

Panel B. Summary statistics

Category	Contract	Mean (%)	SD (%)	SR	Skew	Kurt	Max	Min
Agriculture	Canola	0.28	21.18	0.01	-0.21	6.15	25.09	-33.38
	Rapeseed	8.14	20.13	0.40	0.03	4.43	22.46	-19.14
	Rubber	-6.43	27.80	-0.23	-0.07	3.48	22.27	-28.59
Metals	Iron Ore	24.26	37.55	0.65	-0.48	2.94	24.14	-27.62

Table A.7: Factor momentum in high- and low-eigenvalue factors – More developed markets

This table summarizes the factor momentum performance across different sets of PC factors for a larger dataset of developed markets. The high- (low-) eigenvalue set includes PC factors with high (low) eigenvalues. Panel A reports monthly descriptive statistics for momentum strategies, including the mean returns (in percent) and their t -statistics. Panels B and C report spanning test results for the full sample and subsample after the financialization. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A. Factor momentum in subsets of PC factors ordered by eigenvalues

Sets of PCs	Full sample			Subsample		
	Mean	t -stat	$t(\alpha_{BP2})$	Mean	t -stat	$t(\alpha_{BP2})$
High	0.41**	2.54	2.52	0.52***	2.71	2.93
Low	0.32*	1.83	1.83	0.34*	1.74	1.70

Panel B. Explaining factor momentum in high-eigenvalue PC factors

	FMOM ^{HighPC}	
	Full sample	Subsample
α	0.18 (1.07)	0.21 (1.07)
FMOM ^{HighPC}	0.35*** (6.68)	0.22*** (3.25)
Controls	Y	Y
N	388	221
Adj. R^2	9.7%	3.8%

Panel C. Explaining factor momentum in low-eigenvalue PC factors

	FMOM ^{LowPC}	
	Full sample	Subsample
α	0.32** (2.03)	0.21*** (2.60)
FMOM ^{LowPC}	0.30*** (6.68)	0.21*** (3.25)
Controls	Y	Y
N	388	221
Adj. R^2	9.9%	5.1%

Table A.8: Time-series efficient pricing models – More developed markets

This table compares the performance of the Boons and Prado (2019) model based on the time-series efficient factors (MKT* and Basis momentum*) and standard factors (MKT and Basis momentum) for the full and subsample for a larger dataset of developed markets. Panel A spans the efficient model by the standard model, whereas Panel B does the opposite. We report the alphas and their t -statistics in parentheses, and the Gibbons et al. (1989) (GRS) test F -statistics and their p -values in square brackets. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

<i>Panel A. Efficient model regressed on standard model</i>		
	Full sample	Subsample
	α_{BP2}	α_{BP2}
MKT*	0.03 (0.40)	0.12 (0.93)
Basis momentum*	0.23*** (2.73)	0.52*** (2.82)
GRS F -statistic	3.74** [0.02]	4.48*** [0.01]
<i>Panel B. Standard model regressed on efficient model</i>		
	Full sample	Subsample
	α_{BP2}^*	α_{BP2}^*
MKT	-0.02 (-0.11)	-0.07 (-0.27)
Basis momentum	-0.13 (-1.42)	-0.19 (-0.78)
GRS F -statistic	1.00 [0.37]	0.33 [0.72]

Table A.9: Commodity futures contract information for developing markets

This table summarizes contract information for 68 commodity futures traded in developing markets. See also notes to Table A.1.

Category	Contract	Exchange	Market	Start Month
Agriculture	Coffee	B3	Brazil	200401
	Common Wheat	ZCE	China	201201
	Corn	DCE	China	200409
	Cornstarch	DCE	China	201412
	Early Rice	ZCE	China	200904
	Egg	DCE	China	201311
	Guar seed	NDX	India	201405
	Hard Wheat	ZCE	China	200401
	Japonica Rice	DCE	China	201311
	Jujube	ZCE	China	201904
	Late Rice	ZCE	China	201407
	Live Hog	DCE	China	202101
	No.1 Soybean	DCE	China	200401
	No.2 Soybean	DCE	China	200412
	Palm Olein	DCE	China	200710
	Peanut Kernel	ZCE	China	202102
	Polished Rice	DCE	China	201908
	Rapeseed	ZCE	China	201212
	Rapeseed Meal	ZCE	China	201212
	Rapeseed Oil	ZCE	China	200706
	Apple	ZCE	China	201712
	Soybean Meal	DCE	China	200401
	Soybean Oil	DCE	China	200601
	Strong Wheat	ZCE	China	200401
	Sugar	ZCE	China	200601
Energy	Crude Oil	SHFE	China	201803
	Fuel Oil	SHFE	China	200408
	Methanol	ZCE	China	201110
	Natural gas	MOEX	Russia	202002
Industrial materials	Bitumen	SHFE	China	201310
	Coke	DCE	China	201104
	Coking Coal	DCE	China	201303
	Cotton	ZCE	China	200406
	Cotton Yarn	ZCE	China	201708
	Ethenylbenzene	DCE	China	201909
	Ethylene Glycol	DCE	China	201812
	Fiberboard	DCE	China	201312
	Flat Glass	ZCE	China	201212
	Hot-Rolled Coil	SHFE	China	201403
	Liquefied Petroleum Gas	DCE	China	202003
	LLDPE	DCE	China	200707
	Low Sulfur Fuel Oil	SHFE	China	202006
	Natural Rubber	SHFE	China	200401
	Plywood	DCE	China	201312
	Polyester Staple Fiber	ZCE	China	202010
	Polypropylene	DCE	China	201402
	PTA	ZCE	China	200612
	PVC	DCE	China	200905
	Soda Ash	ZCE	China	201912
	Softwood Kraft Pulp	SHFE	China	201811
	Thermal Coal	ZCE	China	201309
	TSR 20	SHFE	China	201908
	Urea	ZCE	China	201908
Metal	Aluminum	SHFE	China	200401
	Copper	SHFE	China	200401
	Copper Cathode	SHFE	China	202011
	Ferrosilicon	ZCE	China	201408
	Gold	SHFE	China	200801
	Iron Ore	DCE	China	201310
	Lead	SHFE	China	201103
	Nickel	SHFE	China	201503
	Silicon Manganese	ZCE	China	201408
	Silver	SHFE	China	201205
	Stainless Steel	SHFE	China	201909
	Steel Rebar	SHFE	China	200903
	Tin	SHFE	China	201503
	Wire Rod	SHFE	China	200903
	Zinc	SHFE	China	200703

Table A.10: Summary statistics of commodity futures returns in developing markets

This table reports the mean (in percent), standard deviation (SD, in percent), Sharpe ratio (SR), skewness (Skew), kurtosis (Kurt), maximum (Max, in percent), and minimum (Min, in percent) of excess futures returns of 68 commodity futures contracts in developing markets.

Category	Contract	Mean	SD	SR	Skew	Kurt	Max	Min
Agriculture	Coffee	-8.33	30.39	-0.27	0.32	3.53	34.71	-24.84
	Common Wheat	-1.01	11.23	-0.09	0.90	5.11	11.09	-6.95
	Corn	0.99	10.24	0.10	-0.15	3.20	8.09	-8.28
	Cornstarch	2.50	14.95	0.17	-0.15	3.30	10.97	-11.88
	Early Rice	-2.90	11.43	-0.25	0.66	5.25	13.21	-9.54
	Egg	-12.97	21.42	-0.61	-0.63	4.48	15.30	-21.17
	Guar seed	-11.87	30.53	-0.39	0.39	4.70	29.88	-25.78
	Hard Wheat	-7.23	8.58	-0.84	-0.99	6.23	4.75	-11.84
	Japonica Rice	-5.58	15.61	-0.36	0.91	6.06	18.53	-10.54
	Jujube	0.06	26.79	0.00	0.86	5.44	24.64	-18.51
	Late Rice	-5.55	15.22	-0.36	-3.01	20.43	7.27	-28.78
	Live Hog	-33.87	32.06	-1.06	-0.51	3.29	13.93	-24.87
	No.1 Soybean	3.36	15.44	0.22	-0.07	4.58	14.40	-18.08
	No.2 Soybean	4.63	16.33	0.28	-0.40	4.24	12.33	-18.55
	Palm Olein	1.64	26.36	0.06	-0.66	6.67	21.92	-37.73
	Peanut Kernel	-5.35	18.72	-0.29	0.06	2.27	9.82	-9.71
	Polished Rice	-8.10	8.62	-0.94	-1.70	9.32	3.80	-10.77
	Rapeseed	1.76	14.10	0.13	-1.57	12.14	10.68	-23.23
	Rapeseed Meal	10.67	19.21	0.56	0.23	2.99	15.63	-12.29
	Rapeseed Oil	1.32	20.48	0.06	-0.53	9.89	29.25	-29.63
	Apple	0.03	29.19	0.00	0.16	3.46	19.48	-23.71
	Soybean Meal	7.23	20.01	0.36	0.11	3.74	18.97	-20.91
	Soybean Oil	2.51	21.08	0.12	-0.54	7.12	25.10	-26.93
	Strong Wheat	-6.39	10.43	-0.61	-0.42	4.61	8.30	-12.48
	Sugar	-2.77	18.91	-0.15	0.52	6.79	26.77	-18.43
Energy	Crude Oil	4.44	35.98	0.12	-1.18	4.79	18.22	-35.17
	Fuel Oil	-2.97	29.56	-0.10	-1.18	9.55	30.83	-50.95
	Methanol	-8.41	26.93	-0.31	-0.70	5.58	22.53	-31.85
	Natural gas	39.09	50.66	0.77	0.04	1.99	27.20	-21.76
Industrial materials	Bitumen	-11.59	31.32	-0.37	-0.77	6.10	20.83	-42.21
	Coke	2.18	32.74	0.07	0.36	3.97	31.20	-25.17
	Coking Coal	10.16	32.67	0.31	0.02	4.15	25.27	-30.17
	Cotton	-4.00	19.60	-0.20	0.38	7.22	25.70	-25.57
	Cotton Yarn	-5.61	18.89	-0.30	0.09	3.15	14.58	-13.25
	Ethylbenzene	11.22	41.47	0.27	-1.11	7.55	29.88	-43.08
	Ethylene Glycol	-8.94	33.72	-0.27	-0.44	5.45	24.36	-33.95
	Fiberboard	-8.88	32.73	-0.27	-1.71	10.02	19.54	-48.54
	Flat Glass	5.79	24.08	0.24	0.11	5.20	27.55	-20.72
	Hot-Rolled Coil	10.97	24.82	0.44	-0.12	3.00	18.85	-20.66
	Liquefied Petroleum Gas	43.24	34.11	1.27	-0.23	3.77	25.77	-21.67
	LLDPE	-2.27	25.92	-0.09	-2.94	24.80	18.13	-60.09
	Low Sulfur Fuel Oil	44.61	28.97	1.54	-0.26	2.32	16.76	-13.38
	Natural Rubber	-12.24	30.73	-0.40	-0.21	4.53	27.35	-40.18
	Plywood	-2.94	31.99	-0.09	0.48	6.67	32.77	-35.65
	Polyester Staple Fiber	17.61	27.76	0.63	0.12	4.64	22.65	-16.80
	Polypropylene	4.95	22.56	0.22	0.42	3.96	19.03	-16.56
	PTA	-5.43	26.91	-0.20	-0.93	9.63	30.03	-45.21
	PVC	1.75	21.11	0.08	-0.01	6.99	26.02	-25.76
	Soda Ash	7.67	36.65	0.21	0.45	3.12	24.18	-19.42
	Softwood Kraft Pulp	1.48	23.58	0.06	0.08	2.94	14.86	-15.31
	Thermal Coal	8.83	27.34	0.32	0.71	10.78	41.90	-30.84
Metal	TSR 20	-5.14	24.67	-0.21	-0.23	3.40	15.67	-17.41
	Urea	21.94	25.57	0.86	-0.01	3.70	20.23	-17.41
	Aluminum	-1.16	16.50	-0.07	-0.13	4.21	14.83	-14.38
	Copper	6.83	26.70	0.26	-1.44	13.57	29.32	-52.83
	Copper Cathode	17.42	20.23	0.86	1.02	4.49	17.74	-8.09
	Ferrosilicon	-2.06	36.34	-0.06	-0.59	10.22	42.97	-43.48
	Gold	1.36	16.85	0.08	-0.51	4.94	12.39	-20.70
	Iron Ore	18.97	35.91	0.53	-0.27	3.04	26.98	-28.30
	Lead	-2.21	17.56	-0.13	0.44	7.68	24.59	-19.14
	Nickel	6.50	24.52	0.27	-0.19	2.49	17.82	-16.72
	Silicon Manganese	9.02	31.38	0.29	0.22	5.86	34.10	-29.00
	Silver	-9.02	24.79	-0.36	0.11	5.13	24.66	-23.16
	Stainless Steel	11.52	23.72	0.49	0.42	2.66	18.37	-9.76
	Steel Rebar	3.73	23.43	0.16	-0.18	3.80	19.13	-23.64
	Tin	10.24	18.96	0.54	-0.38	4.07	12.90	-18.51
	Wire Rod	4.06	20.09	0.20	0.52	5.89	25.28	-16.92
	Zinc	-3.20	24.76	-0.13	-1.87	12.77	17.94	-46.77

Table A.11: Factor momentum in high- and low-eigenvalue factors – Developing markets

This table summarizes the factor momentum performance across different sets of PC factors for developing markets. The high- (low-) eigenvalue set includes PC factors with high (low) eigenvalues. Panel A reports monthly descriptive statistics for momentum strategies, including the mean returns (in percent) and their t -statistics. Panels B and C report spanning tests. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

<i>Panel A. Factor momentum in subsets of PC factors ordered by eigenvalues</i>			
	Mean	<i>t</i> -stat	<i>t</i> (α_{BP2})
High	0.41***	2.68	2.12
Low	0.34**	1.97	1.39
<i>Panel B. Explaining factor momentum in low-eigenvalue PC factors</i>			
	FMOM ^{LowPC}		
α	0.22 (1.22)		
FMOM ^{HighPC}	0.09 (1.07)		
Controls	Y		
N	208		
Adj. <i>R</i> ²	3.0%		
<i>Panel C. Explaining factor momentum in high-eigenvalue PC factors</i>			
	FMOM ^{HighPC}		
α	0.32** (2.01)		
FMOM ^{LowPC}	0.18 (1.07)		
Controls	Y		
N	208		
Adj. <i>R</i> ²	4.4%		

Table A.12: Time-series efficient pricing models – Developing markets

This table compares the performance of the Boons and Prado (2019) model based on the time-series efficient factors (MKT* and Basis momentum*) and standard factors (MKT and Basis momentum) for developing markets. Panel A spans the efficient model by the standard model, whereas Panel B does the opposite. We report the alphas and their t -statistics in parentheses, and the Gibbons et al. (1989) (GRS) test F -statistics and their p -values in square brackets. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

<i>Panel A. Efficient model regressed on standard model</i>	
	α_{BP2}
MKT*	0.01 (0.04)
Basis momentum*	0.31*** (3.55)
GRS F -statistic	7.16*** [0.00]
<i>Panel B. Standard model regressed on efficient model</i>	
	α_{BP2}^*
MKT	-0.23 (-0.99)
Basis momentum	-0.21** (-1.98)
GRS F -statistic	2.13 [0.12]